

Integral Homology of Singleton-Anchor Configuration Spaces of Finite Graphs

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Abstract

Let G be a nonempty finite graph, let k be a vertex, and let $n \geq 1$. We determine the integral homology of the space of ordered n -tuples in G^n for which at least one coordinate equals k . If c is the number of components of G and $\beta = \text{rank } H_1(G; \mathbb{Z})$, then its q th Betti number is

$$\binom{n}{q} \beta^q (c^{n-q} - (c-1)^{n-q}).$$

All homology groups are free abelian. The proof reduces the pointed graph to a disjoint union of bouquets and computes the cellular chains of the resulting fat wedge. Consequently, a torsion example among anchored configuration spaces, if one exists, must use at least two anchors and at least two more coordinates than anchors. In particular, singleton anchors are eliminated from the cograph torsion problem, but the multiple-anchor problem remains unresolved.

1 Context and statement

For a finite graph G , a set of vertices K , and an integer $n \geq |K|$, the anchored configuration space is

$$\Sigma(G, K, n) = \{(x_1, \dots, x_n) \in G^n : K \subseteq \{x_1, \dots, x_n\}\}.$$

The coordinates are labeled, collisions are allowed, and G is regarded as a one-dimensional CW complex. These spaces and their generalized variants have been studied in particular through Euler characteristics and through complete homology calculations for circle graphs [1]. For cographs, Mamun, Nalikka, and Ramos proved a fixed-parameter finite-generation theorem and derived restrictions on possible integral torsion [2, Corollary 4.18]. Their subsequent discussion records that torsion is not known in general outside the established tree and cycle cases.

Here we treat the singleton slice $K = \{k\}$ without imposing any cograph or connectivity hypothesis. Write

$$c = \text{rank } H_0(G; \mathbb{Z}), \quad \beta = \text{rank } H_1(G; \mathbb{Z}) = |E(G)| - |V(G)| + c.$$

We use the convention $a^0 = 1$, including $0^0 = 1$.

Theorem 1. *Let G be a nonempty finite graph, possibly disconnected, let $k \in V(G)$, and let $n \geq 1$. For $0 \leq q \leq n$,*

$$H_q(\Sigma(G, \{k\}, n); \mathbb{Z}) \cong \mathbb{Z}^{b_q}, \quad b_q = \binom{n}{q} \beta^q (c^{n-q} - (c-1)^{n-q}).$$

In particular $b_n = 0$, all groups in degrees greater than n vanish, and every integral homology group is torsion-free.

2 Fat-wedge reduction and proof

For a based space (X, x_0) , define its n th fat wedge by

$$F_n(X, x_0) = \{(x_1, \dots, x_n) \in X^n : x_i = x_0 \text{ for some } i\}.$$

This construction preserves based homotopy equivalences. Indeed, a based map $f: (X, x_0) \rightarrow (Y, y_0)$ induces the coordinatewise map $F_n(f)$, and a based homotopy H induces a coordinatewise homotopy on fat wedges: a coordinate equal to x_0 remains equal to y_0 throughout the homotopy. Applying this observation to based homotopy inverses proves the assertion.

Choose a spanning tree in each component of G and collapse every chosen tree to a vertex. Root the tree in the component containing k at k ; contracting its edges toward the root gives contractions and homotopies that fix k . The resulting quotient is a based homotopy equivalence

$$(G, k) \simeq (W, *),$$

where W is a disjoint union of c bouquets of circles, $*$ is the bouquet vertex corresponding to the component of k , and the bouquets contain β loop one-cells in total. Therefore

$$\Sigma(G, \{k\}, n) = F_n(G, k) \simeq F_n(W, *).$$

Give W^n the product CW structure. The fat wedge $F_n(W, *)$ is precisely the subcomplex formed by product cells having at least one factor equal to the zero-cell $*$. Every one-cell of W is a loop and hence has cellular boundary zero. The product boundary formula then shows that every cellular differential of $F_n(W, *)$ is zero. Its integral homology is consequently its free cellular chain group.

It remains only to count cells. A q -cell is specified by choosing the q coordinates occupied by loop cells, choosing one of the β loop cells in each of those coordinates, and choosing zero-cells in the remaining $n - q$ coordinates with at least one choice equal to $*$. The number of choices is

$$\binom{n}{q} \beta^q (c^{n-q} - (c-1)^{n-q}).$$

When $q = n$, there is no zero-cell coordinate and hence no cell in the fat wedge; equivalently, the final factor in the formula is $1 - 1 = 0$. This proves Theorem 1.

3 Consequences for torsion searches

Corollary 2. *Let G be a nonempty finite graph, let $K \subseteq V(G)$, and let $n \geq |K|$. If the integral homology of $\Sigma(G, K, n)$ has nonzero torsion, then*

$$|K| \geq 2 \quad \text{and} \quad n \geq |K| + 2.$$

The same necessary conditions hold when G is restricted to be a cograph.

Proof. If K is empty, then $\Sigma(G, K, n) = G^n$, whose integral homology is free by the Künneth theorem because the homology of a graph is free. The singleton case is Theorem 1. Now let $r = |K|$. A product cell contained in $\Sigma(G, K, n)$ must have at least r distinct coordinates occupied by the distinct anchor vertices. It therefore has dimension at most $n - r$. If $n \leq r + 1$, the space is at most one-dimensional, so its integral homology is torsion-free. Thus torsion requires both displayed inequalities. $\square$

The theorem gives all degrees and also covers disconnected graphs. This is not supplied by the closest inspected results: Kozlov gives an Euler characteristic formula for arbitrary connected non-tree graphs and determines the homology of generalized anchored spaces on circle graphs [1], whereas the cograph theorem of Mamun–Nalikka–Ramos is a finite-generation result [2]. As a compatibility check, when G is connected the present formula gives

$$\chi(\Sigma(G, \{k\}, n)) = \sum_{q=0}^{n-1} (-1)^q \binom{n}{q} \beta^q = (1 - \beta)^n - (-\beta)^n,$$

which is the singleton-anchor specialization of the Euler-characteristic formula in [1]. This comparison is bounded to the cited sources; no claim of priority is made.

As a nonessential exact check, take the diamond graph $K_4 - e$. It has $c = 1$ and $\beta = 2$, so Theorem 1 gives Betti numbers $(b_0, b_1, b_2) = (1, 6, 12)$ at $n = 3$. A retained exact cubical computation for representatives of both vertex orbits constructs chain groups of ranks $(37, 105, 75)$, verifies $\partial_1 \partial_2 = 0$, and reduces the relation matrix over $\mathbb{Z}$ to 63 nonzero Smith factors, all equal to one. It therefore recovers the same homology and no torsion. The computation is a consistency check, not an ingredient in the proof.

Computational and AI assistance

An AI system assisted with proof exploration, literature organization, manuscript drafting, and consistency checks. The diamond calculation was performed by a retained Python script using exact integer arithmetic; no floating-point output is used as mathematical evidence. The general theorem is established by the self-contained argument above and does not depend on the computation. The author reviewed the mathematical statements and takes responsibility for the manuscript.

References

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- [2] Adityo Mamun, Jonathan Nalikka, and Eric Ramos. Universality in the algebra and topology of cographs, 2026. arXiv:2609.04554v1.